

# Parthiv Farazi

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## EDUCATION

### Georgia Institute of Technology

Bachelor of Science in Computer Science

Atlanta, GA

Expected Graduation: December 2026

- **Threads:** Intelligence + Modeling & Simulation
- **Coursework:** Machine Learning, Artificial Intelligence, Algorithm Design and Analysis, Data Structures, Software Engineering, Operating Systems, Linear Algebra

## TECHNICAL SKILLS

**Computer Vision:** object detection, keypoint/pose estimation, homography and camera calibration, RANSAC, OpenCV, YOLOv8, PyTorch, torchvision

**AI / ML / LLM:** eval frameworks, Retrieval-Augmented Generation (RAG), Transformer embeddings, FAISS-GPU, EmbeddingGemma, Hugging Face

**Languages:** Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML/CSS

**Backend / Infra:** Supabase (PostgreSQL), FastAPI, Cloudflare R2, Docker, GitHub Actions, GCP, Slurm HPC, CUDA, REST, Git, Linux

**Frameworks:** React, React Native, Expo, Next.js, Three.js, Framer Motion, Tailwind CSS, Django

## EXPERIENCE

### UPDT. (Update Analytics)

Atlanta, GA

Co-founder & CTO

Jan 2026 – Present

- Co-founded **UPDT.** (updt.pro), an AI soccer analytics platform that **turns wide-angle match footage into scouting data**, with a UEFA B-licensed analyst, backed by Georgia Tech's **Create-X**. Own **CV/ML, infrastructure, and product end-to-end**, building directly with the analyst as the first user.
- Built the **computer-vision system that turns raw wide-angle match video (Veo camera) into player and ball tracking**. When one combined model could not reliably detect the ball, diagnosed the failure and split it into a **two-model design that fixed it**, trained on GT's PACE-ICE GPU cluster (24 clips, 480 frames, ball detection mAP50 0.42).
- Built a **quality-check system that catches bad outputs the overall accuracy score hides**, flagging errors before they reach users by grading every output GOOD/MEDIUM/BAD/FAILED on three signals with a failure-mode taxonomy (3,097 frames at 1.4-px mean reprojection error).

### Qard (Fintech Startup)

Atlanta, GA

Founding Frontend Developer

Jun 2025 – Aug 2025

- Grew an early-stage fintech startup's launch to **200+ users in under a month** by designing and building its production landing page end-to-end in **React + TypeScript**.
- Lifted **average session duration by 40%** with an interactive 3D card system (Three.js + Framer Motion), and cut initial page load by **35%** across devices via dynamic imports.

### RMAICT International

Kuala Lumpur, Malaysia

AI Engineer Intern

May 2024 – Aug 2024

- Cut a team's **manual data entry by 3 hours/week** by deploying a **Document Understanding Transformer (Donut)** via Hugging Face that converts raw receipt images into structured JSON, fine-tuned on **1,000+ receipts**.
- Saved **another 2 hours/week** by extending the model through **transfer learning** to invoices and other financial documents.

## PROJECTS

### Embedding-Based Quote Retrieval (Production RAG Pipeline) | FAISS-GPU, EmbeddingGemma Nov 2025

- Built a semantic search and **Retrieval-Augmented Generation (RAG)** system over **493k documents** at **sub-100ms** nearest-neighbor latency, using FAISS-GPU and Google's EmbeddingGemma-300m (768-dim vectors).
- Built the full 493k index in **25 minutes** on Georgia Tech's PACE GPU cluster (Slurm batch jobs, CUDA memory tuning), choosing **FAISS-GPU over pgvector** after weighing query latency against update cadence.

### Pong Baseball Automation App | React Native, TypeScript, Supabase (live on App Store) Nov 2025 – Jan 2026

- Designed, built, and shipped solo end-to-end (frontend, backend, auth, database, deployment, and **Apple App Store review**), replacing a paper-based stat-tracking system now used by **70+ live users** with real-time tracking and cloud sync.
- Eliminated **2+ hours of post-game manual data entry per game** with a scalable Supabase (PostgreSQL, Auth, Storage) backend.